

Seminar 14 - Algebră

Vectori și valori proprii

- Fie V K -sp. vect., K -corp comutativ ($K \in \{\mathbb{R}, \mathbb{C}\}$), $\dim V = n$
- Fie $f \in \text{End}_K(V)$
- Fie $x \in V \setminus \{0\}$. Dacă $\exists \lambda \in K$ a.c. $f(x) = \lambda x$, atunci x s.m. vector propriu a lui f , iar λ s.m. val. proprie a lui f (resp. lui x)
- Fie $\lambda \in K$ o val. proprie a lui f
 $V(\lambda) = \{x \in V : f(x) = \lambda x\} = \text{Ker}(f - \lambda \cdot \text{Id}_V)$ s.m. subspațiul propriu coresp. lui λ
- Fie $B = (v_1, v_2, \dots, v_n)$ bază p. V
- Not $P_f(\lambda) = \det(fB - \lambda I_n)$ pol. caracter. al lui f
- Val. proprii ale lui f sunt soluțiile ec. $P_f(\lambda) = 0$

ex: a) Det. vectorii și val. proprii ale matricei

$$A = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \in M_3(\mathbb{R})$$

Formula

b) Ar. c.c. $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$

$$(A - \lambda I) v = 0$$

$f(x_1, x_2, x_3) = (2x_1 - 2x_3, 3x_2, 3x_3)$ e un endomorfism al lui $\mathbb{R}^3$ și det. mat. și val. proprii

Sol: a) $P_A(\lambda) = \det(A - \lambda I_3) = \begin{vmatrix} 2-\lambda & 0 & -2 \\ 0 & 3-\lambda & 0 \\ 0 & 0 & 3-\lambda \end{vmatrix} = (2-\lambda)(3-\lambda)^2$

$$P_A(\lambda) = 0 \rightarrow \lambda_1 = 2$$

$$\lambda_2 = \lambda_3 = 3$$

$$V(2) = \{ (x_1, x_2, x_3) \in \mathbb{R}^3 : (A - 2I_3) \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \}$$

$$(A - \lambda I_3) \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Leftrightarrow \begin{cases} -2x_3 = 0 \\ x_2 = 0 \\ x_3 = 0 \end{cases}$$

$$\Rightarrow V(2) = \{ (x_1, 0, 0) : x_1 \in \mathbb{R} \}$$

$$V(3) = \{ (x_1, x_2, x_3) \in \mathbb{R}^3 : (A - \lambda I_3) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \}$$

$$(A - \lambda I_3) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Leftrightarrow \begin{cases} -x_1 = 2x_3 = 0 \end{cases}$$

$$\Rightarrow V(3) = \{ (-2x_3, x_2, x_3) : x_2, x_3 \in \mathbb{R} \}$$

$$\begin{aligned} \text{Ket proprii ai lui } A &= V(2) \cup V(3) \setminus \{0, 0, 0\} \\ &= \{ (x_1, 0, 0) : x_1 \in \mathbb{R}^* \} \cup \{ (-2x_3, x_2, x_3) : \\ &\quad x_2^2 + x_3^2 \neq 0 \} \end{aligned}$$

$$e) f \text{ transformare lin.} \Rightarrow f \in \text{End}(\mathbb{R}^3)$$

$$f(e_1) = (2, 0, 0)$$

$$f(e_2) = (0, 3, 0)$$

$$f(e_3) = (-2, 0, 3)$$

$$\Rightarrow [f]_e = A$$

• Fie $\lambda_* \in K$ o val proprie a lui —

— multiplicitatea algebrică a lui λ_* = mult răd λ_* în $p_f(\lambda)$

— multiplicitatea geometrică a lui λ_* = $\dim V(\lambda_*)$

$$\dim V(\lambda_*) = \dim \ker(f - \lambda_* \cdot \text{Id}_V)$$

$$= \dim V - \dim \text{Im}(f - \lambda_* \cdot \text{Id}_V)$$

$$= n - \text{rang}([f]_V - \lambda_* \cdot I_n)$$

$$-1 \leq \text{mult. geom. } \lambda_* \leq \text{mult. alg. } \lambda_*$$

$$\text{Obs: } V(2) = \{ x_1(1, 0, 0) : x_1 \in \mathbb{R} \} = \langle (1, 0, 0) \rangle ; \text{mult alg } 2 = 1 = \text{mult geom}$$

$$V(3) = \{ x_2(0, 1, 0) + x_3(-2, 0, 1) : x_2, x_3 \in \mathbb{R} \} = \langle (0, 1, 0), (-2, 0, 1) \rangle$$

$$\text{mult alg } 3 = 2 = \text{mult geom } 3$$

• f nm. diagonalizabil dacă J o bază e o lni V pt care $\{fJ\}$ este o matrice diagonală

• f este diagonalizabil

$\begin{cases} P_f(\lambda) \text{ are toate răd în } K \\ \text{pt orice } \lambda \text{ răd o lni } P_f(\lambda) \\ \text{mult}_{\text{alg}} \lambda = \text{mult}_{\text{geom}} \lambda \end{cases}$

ex: Care dintre urm endomorfisme sunt diagonalizabile

a) $f \in \text{End}_{\mathbb{R}}(\mathbb{R}^3) : f(x, y, z) = (-z, -x, -y)$

b) $f \in \text{End}_{\mathbb{C}}(\mathbb{C}^3) \quad \text{--- " ---}$

c) $f \in \text{End}_{\mathbb{R}}(\mathbb{R}^3) : f(x, y, z) = (-2y - 3z, x + 3y + 3z, z)$

d) $f \in \text{End}_{\mathbb{R}}(\mathbb{R}^4) : f(x_1, x_2, x_3, x_4) = (-2x_1, -2x_2, 3x_3, x_3 + 3x_4)$

Sol:

$$P_f(\lambda) = \det(fJ - \lambda J_3)$$

For (e_1, e_2, e_3) bază canonică în $\mathbb{R}^3$

$$f(e_1) = (0, -1, 0)$$

$$f(e_2) = (0, 0, -1)$$

$$f(e_3) = (-1, 0, 0)$$

$$\Rightarrow [f]_e = \begin{bmatrix} 0 & 0 & -1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}$$

$$\begin{vmatrix} -\lambda & 0 & -1 \\ -1 & -\lambda & 0 \\ 0 & -1 & -\lambda \end{vmatrix} = (\lambda + 1)(\lambda^2 - \lambda + 1)$$

$$P_f(\lambda) = 0 \Rightarrow \lambda_1 = -1$$

$$\lambda_{2,3} \in \mathbb{C} \setminus \mathbb{R} \Rightarrow f \text{ nu e diag}$$

b) $\lambda_1 = -1, \lambda_{2,3} \in \mathbb{C} \setminus \mathbb{R}$

$$\text{mult}_{\text{alg}} \lambda_1, \lambda_2, \lambda_3 = 1 = \text{mult}_{\text{geom}} \quad (\text{răd simple})$$

$$\Rightarrow f \text{ e diagonalizabilă}$$

c) $T_{\mathbb{R}}(e_1, e_2, e_3) = \text{basis canonisch } \mathbb{R}^3$

$$\begin{aligned} f(e_1) &= (0, 1, 0) \\ f(e_2) &= (-2, 3, 0) \\ f(e_3) &= (-3, 3, 1) \end{aligned} \Rightarrow \{f\}_{\mathcal{B}} = \begin{bmatrix} 0 & -2 & -3 \\ 1 & 3 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P_f(\lambda) = \det(\{f\}_{\mathcal{B}} - \lambda I_3) = \begin{vmatrix} -\lambda & -2 & -3 \\ 1 & 3-\lambda & 3 \\ 0 & 0 & 1-\lambda \end{vmatrix} = (1-\lambda)(\lambda^3 - 3\lambda + 2) \\ = (1-\lambda)(\lambda-2)^2 = (1-\lambda)(\lambda-2)$$

$$P_f(\lambda) = 0 \Rightarrow \lambda_1 = \lambda_2 = 1 \quad e_{\mathbb{R}} \quad (2 = \text{mult alg } 1) \\ \lambda_3 = 2$$

Mod sample (1 = mult alg, 2 = mult geom)

$$\begin{aligned} \dim V(1) &= 3 - \text{rang}(\{f\}_{\mathcal{B}} - 1I_3) \\ &= 3 - \text{rang} \begin{bmatrix} -1 & -2 & -3 \\ 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix} \\ &= 2 \Rightarrow \text{mult geom } 1 = 2 = \text{mult alg } 1 \end{aligned}$$

Resultat: f diagonalisierbar

d) $T_{\mathbb{R}}(e_1, e_2, e_3, e_4) = \mathcal{B}$ basis canonisch in $\mathbb{R}^4$

$$\begin{aligned} f(e_1) &= (-2, 0, 0, 0) \\ f(e_2) &= (0, -2, 0, 0) \\ f(e_3) &= (0, 0, 3, 1) \\ f(e_4) &= (0, 0, 0, 3) \end{aligned} \Rightarrow \{f\}_{\mathcal{B}} = \begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$$\begin{aligned} P_f(\lambda) &= \det(\{f\}_{\mathcal{B}} - \lambda I_4) = \begin{vmatrix} -2-\lambda & 0 & 0 & 0 \\ 0 & -2-\lambda & 0 & 0 \\ 0 & 0 & 3-\lambda & 0 \\ 0 & 0 & 1 & 3-\lambda \end{vmatrix} \\ &= (3-\lambda)^2 (2+\lambda)^2 \end{aligned}$$

$$P_f(\lambda) = 0 \Rightarrow \begin{cases} \lambda_1 = \lambda_2 = 3 \\ \lambda_3 = \lambda_4 = -2 \end{cases} e_{\mathbb{R}}$$

$$\dim V(3) = 4 - \text{rang}(\lambda I_4 - 3I_4) =$$

$$= 4 - \text{rang} \begin{bmatrix} -5 & 0 & 0 & 0 \\ 0 & -5 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = 4 - 3 = 1$$

$$\Rightarrow \text{mult } \lambda_3 = 2 \neq 1 = \text{mult geom } 3$$

$\Rightarrow f$ nicht diagonalisierbar

ex: Für $A = \begin{bmatrix} 0 & -2 & -3 \\ 1 & 3 & 3 \\ 0 & 0 & 1 \end{bmatrix} \in M_3(\mathbb{R})$, $\lambda I_3 = A$

a) A diagonalisierbar?

b) Gibt es mit $S \in GL_3(\mathbb{R})$ so $S^{-1}AS$ ist diagonalisierbar?

c) $A^m = ?$ $m \in \mathbb{N}^+$

Sol: a) A ist diagonalisierbar

$$\text{vel proprii} = \begin{matrix} 1 & (\text{mult } \lambda_2) \\ 2 & (\text{mult } \lambda_1) \end{matrix}$$

$$V(1) = \left\{ (x_1, x_2, x_3) \in \mathbb{R}^3 : (A - 1 \cdot I_3) \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\}$$

$$= \left\{ (-2x_2 - 3x_3, x_2, x_3) : x_2, x_3 \in \mathbb{R} \right\}$$

$$(A - 1 \cdot I_3) \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Leftrightarrow \begin{cases} -x_1 - 2x_2 - 3x_3 = 0 \\ x_1 + 2x_2 + 3x_3 = 0 \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} x_1 = -2x_2 - 3x_3 \\ x_2 \in \mathbb{R} \\ x_3 \in \mathbb{R} \end{cases}$$

$$= \left\{ x_2(-2, 1, 0) + x_3(-3, 0, 1) : x_2, x_3 \in \mathbb{R} \right\}$$

$$= \left\{ \underbrace{(-2, 1, 0)}_{v_1}, \underbrace{(-3, 0, 1)}_{v_2} \right\}$$

$\Rightarrow v_1, v_2$ vect. proprii lin indep basis vel proprii

$$V(2) = \left\{ (x_1, x_2, x_3) \in \mathbb{R}^3 : (A - 2I_3) \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\}$$

$$(A - 2I_3) \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Leftrightarrow \begin{cases} -2x_1 - 2x_2 - 3x_3 = 0 \\ x_1 + x_2 + 3x_3 = 0 \\ -3x_3 = 0 \end{cases} \Leftrightarrow \begin{cases} x_1 = -x_2 \\ x_2 \in \mathbb{R} \\ x_3 = 0 \end{cases}$$

$$\Rightarrow V(2) = \{ (-x_2, x_2, 0) = x_2 \in \mathbb{R} \} = \{ x_2 \langle -1, 1, 0 \rangle = x_2 \in \mathbb{R} \} \\ = \underbrace{\langle -1, 1, 0 \rangle}_{v_3}$$

$\Rightarrow v = (v_1, v_2, v_3)$ basis in $\mathbb{R}^3$ forming the basis for the eigenspace of A

$$f(v_1) = L \cdot v_1$$

$$f(v_2) = L \cdot v_2$$

$$f(v_3)$$

$$\Rightarrow [f]_v = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$v = e \cdot S ; S = \begin{bmatrix} -2 & -3 & -1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\begin{array}{ccc} \mathbb{R}^3 : & e & \xrightarrow{S} & v \\ A = [f]_e & \downarrow & & \downarrow & [f]_v \\ \mathbb{R}^3 : & e & \xrightarrow{S} & v & \end{array}$$

$\xleftarrow{S^{-1}}$

$$B = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} = S^{-1}AS$$

$$c) B = S^{-1}AS \Rightarrow A = SBS^{-1} \quad |^m$$

$$\Rightarrow A^m = \underbrace{(SBS^{-1})(SBS^{-1}) \dots (SBS^{-1})}_{m \text{ factors}}$$

$$= SB(\underbrace{S^{-1}S}_{I}) \cdot B(S^{-1}S) \dots (S^{-1}S)BS^{-1} \\ = SB^mS^{-1} =$$

$$= S \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2^n \end{pmatrix} S^{-1}$$

$$Q_0 = 0$$

$$Q_1 = 1$$

$$Q_m = Q_{m-1} + Q_{m-2}, \quad m \geq 2$$

$$\text{Ans: } Q_m = \frac{\varphi^m - \psi^m}{\sqrt{5}}, \quad m \geq 0$$

$$\varphi = \frac{1+\sqrt{5}}{2}, \quad \psi = \frac{1-\sqrt{5}}{2}$$

Inductive

$$\begin{pmatrix} Q_m \\ Q_{m+1} \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}}_A \cdot \begin{pmatrix} Q_{m-1} \\ Q_{m-2} \end{pmatrix}$$

$$\stackrel{\text{ind}}{=} A^{m-1} \begin{pmatrix} Q_1 \\ Q_0 \end{pmatrix}$$